



King Abdul Aziz University

Faculty of Computing & Information Technology
Computer Science Department
Spring 2026

CPCS222 (Discrete Structures)

Assignment# 2

Hand-in Date: Sunday 03-05- 2026

Due Date: Thursday 14-05- 2026

Homework Material (Chapter 2)

CLOS:

1. Use set notation to express sets and represent sets graphically using Venn diagrams.
2. Calculate the union, intersection, complement, cardinality, power set, and Cartesian product of sets.
3. Determine the equality of sets using membership tables and set identities.
4. Determine the domain, co-domain, range, type, inverse, and composition of functions.
5. List and produce the terms of a sequence and find its formulae.
6. Express the sum of a sequence and compute its value.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Total
6	9	9	9	4	6	6	9	15	6	6	15	100

Q1. Draw the Venn diagrams for each of these combinations of the sets: **(6 pts, 3 pts each)**

- a. $A - (B \cap C)$
- b. $(A \cup B \cup C) \cap D$

Q2. Write down the elements of the following sets: **(9 pts, 3 pts each)**

- a. set A, where set A contains whole numbers between 5 and 40 all divisible by 4.
- b. set B, where set B includes integers from -15 to 25, all multiples of 5.
- c. set C, where set C consists of prime numbers between 20 and 40.

Q3. Let $R = \{x: x = 3n, n \in \mathbb{N}, n \leq 5\}$, $S = \{x \in \mathbb{N}: x \leq 12\}$ Find:

(9 pts, 3 pts each)

- a. $R \cup S$
- b. $R \cap S$
- c. $S - R$

Q4. Let $Q = \{x \in \mathbb{N}: 2 \leq x \leq 5\}$

(9 pts, 3 pts each)

- a. Write elements of set Q
- b. Find $P(Q)$
- c. Find $|P(Q)|$

Q5. Compute the Cartesian product $C \times A \times B$ where

(4 pts)

$$A = \{x, y, z\}, B = \{1, 2\}, \text{ and } C = \{a, b\}.$$

Q6. Prove using a membership table that:

(6 pts)

$$A \oplus B = (A \cap B') \cup (A' \cap B)$$

Q7. Use set identities to prove:

(6 pts)

$$A - (B \cup C) = (A - B) \cap (A - C)$$

Q8. Find the inverse of each function (if it exists). If the inverse does not exist, explain why.

(9 pts, 3 pts each)

- a. $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 4x + 9.$

- b. $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = (x - 3) / 5.$
- c. $h : \mathbb{R} \rightarrow \mathbb{R}, h(x) = x^2 + 1.$

Q9. Given $f(x) = 2x + 5$ and $g(x) = x^2 - 3$, both with domain $\mathbb{R}$. Compute the following compositions. **(15 pts, 5 pts each)**

- a. $(f \circ g)(x)$ and then evaluate $(f \circ g)(4).$
- b. $(f \circ f)(x)$ and then evaluate $(f \circ f)(1).$
- c. $(g \circ g)(x)$ and then evaluate $(g \circ g)(0).$

Q10. Find an explicit formula for the n -th term of each sequence below.

(6 pts, 3 pts each)

- a. 1, 4, 9, 16, 25, 36, ...
- b. -2, 4, -8, 16, -32, 64, ...

Q11. List the first 6 terms of each of the following sequences. **(6 pts, 3 pts each)**

- a. $\{a_n\}$ where $a_n = 4n - 3$, for $n \geq 1$.
- b. $\{b_n\}$ where $b_n = 2 \cdot 3^{n-1}$, for $n \geq 1$.

Q12. Compute the value of: **(15 pts, 5 pts each)**

- a. $\sum_{k=1}^6 2k + 5$
- b. $\sum_{n=0}^8 6^n$
- c. $\sum_{x=1}^4 \sum_{y=1}^5 (4x + 6y)$